Supplementary material for "CertiVar: Prevalence-Free Certified Assay Likelihood Ratios for Missense Variant-Effect Predictor Calls"

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Section, equation, figure and table numbers without the prefix S refer to the main paper. Contents: S1 proofs of Propositions 1–3, Theorems 1–2 and Corollary 1; S2 simulation (Fig. S1) and why marginal thresholds do not control the minority class; S3 calibration size, region and gene transfer (Fig. S2A–C), per-gene coverage (Table S1) and prior estimation (Table S2); S4 strata, assay-label sensitivity (Table S3) and calls judged against ClinVar labels (Table S4); S5 implementation.

S1 Proofs

Assumptions, propositions, theorems and the corollary are stated in Section 3.1; Q, C, $\hat{t}_k$, L^+ and U^- are defined in Section 3.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [1, 2]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\ 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta$ by the choice of j_k [3]. If F_k has atoms, write $r = F_k^{-1}(U)$ with U uniform and F_k^{-1} the generalized inverse; then $r^{(k)}_{(j)} = F_k^{-1}(U_{(j)})$ and $F_k(F_k^{-1}(u)) \ge u$, so $F_k(r^{(k)}_{(j)}) \ge U_{(j)}$ and the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Proposition 3. Fix $\mathcal{A}$. Both ranks in (1) are non-increasing in α (the in-expectation rank because $\lceil (n_k + 1)(1 - \alpha) \rceil$ is, the PAC rank because $F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha)$ decreases in α for each j, so the smallest qualifying j cannot increase), so $\hat{t}_k(\alpha)$ is non-increasing in α and the set $\{r_k \le \hat{t}_k(\alpha)\}$ shrinks as α grows. Hence, for any $\hat{\alpha} \in G$ that depends on $\mathcal{B}$ but not on the test variant, $\Pr(k \in C_{\hat{\alpha}} \mid Y = k, \mathcal{A}, \mathcal{B}) \ge \Pr(k \in C_{\max G} \mid Y = k, \mathcal{A})$, and taking expectations over $\mathcal{A}$ and Proposition 1 at $\alpha_k = \max G$ give the first claim. For the PAC rank at level $\delta/|G|$, Proposition 2 applied to each $\alpha \in G$ and a union bound give, with probability at least $1 - \delta$, coverage at least $1 - \alpha$ for every $\alpha \in G$ simultaneously, in particular for $\hat{\alpha}$. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is $\mathrm{Binomial}(n_1, \mathrm{TP}(\alpha))$ with $\mathrm{TP}(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [4] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which

never fail. For LR^+ there are $2|G|$ such bounds (ℓ_{TP} and u_{FP} for each α), so by the union bound all of them hold simultaneously with probability at least $1-\delta$. On that event $\mathrm{TP}(\alpha) \geq \ell_{\mathrm{TP}}(\alpha)$ and $\mathrm{FP}(\alpha) \leq u_{\mathrm{FP}}(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\mathrm{FP}(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The argument for LR^- is identical with u_{FN} and ℓ_{TN}. The event does not involve the levels τ, so it covers every evidence level reported. □

Theorem 2. Fix $\mathcal{A}$, so the thresholds and the hypotheses $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ are fixed. If H_α is true and both endpoints $\ell_{\mathrm{TP}}(\alpha)$ and $u_{\mathrm{FP}}(\alpha)$ at level $\delta/2$ hold, then $\ell_{\mathrm{TP}} \leq \mathrm{TP} \leq \tau\,\mathrm{FP} \leq \tau\, u_{\mathrm{FP}}$, so H_α is not rejected; hence a true H_α is rejected with probability at most δ. Let α^* be the first α in the fixed order for which H_α is true (if none, every claim is true). The procedure claims the level at a false α only if it rejects every hypothesis up to that α, which includes H_{α^*}; this has probability at most δ. The selected α is one of the rejected ones, so the claimed level holds with probability at least $1-\delta$. The functional case is identical with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$. □

Corollary 1. Every quantity bounded above is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is, so Propositions 1–3 and Theorems 1–2 transfer. The two call sets are disjoint for any $\alpha^+, \alpha^- \in G$: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1-q \leq \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \leq \hat{t}_0(\alpha^-)$ and $1-q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \geq \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \leq \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them to the EM estimate of the share [5] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (Section 4.2).

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1-\pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1-\alpha$ while marginal coverage is still near $1-\alpha$. Reweighting to a known test prior [5] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

S3 Calibration size, region and gene transfer, and per-gene results

Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2. When the calibration and test halves of a gene share their LoF share, label-shift-weighted conformal prediction [5] with the true share nearly coincides with pooled conformal; with the EM estimate it inherits the estimation error, which collapsed its LoF coverage in some pairs (worst-class coverage as low as 0.039 at $\alpha = 0.1$).

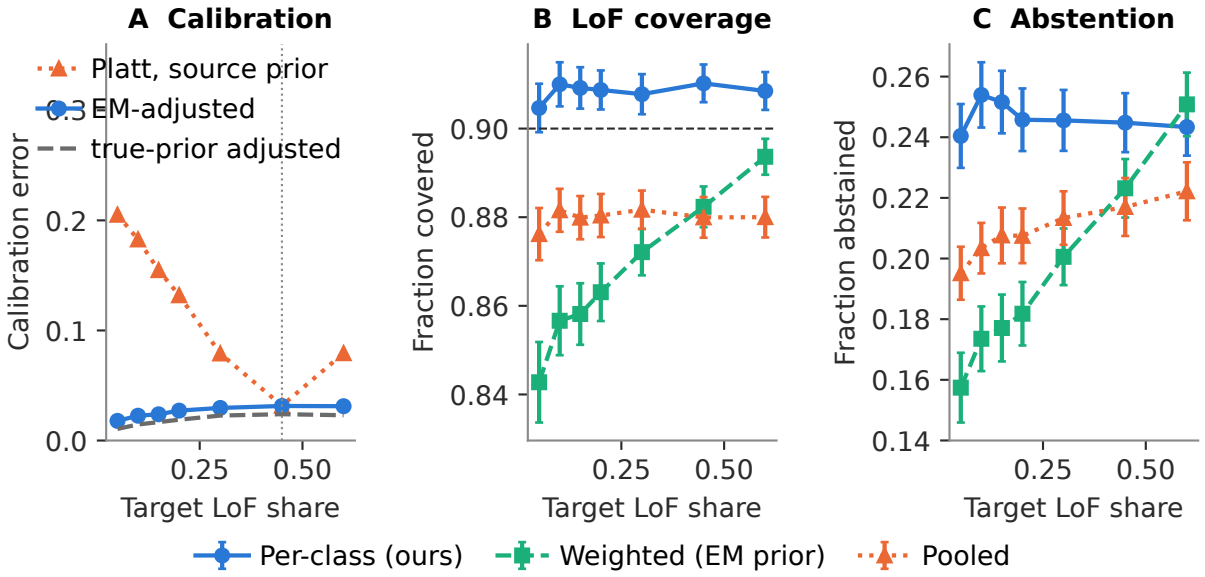


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share. (B) Coverage of LoF variants. (C) Abstention rate.

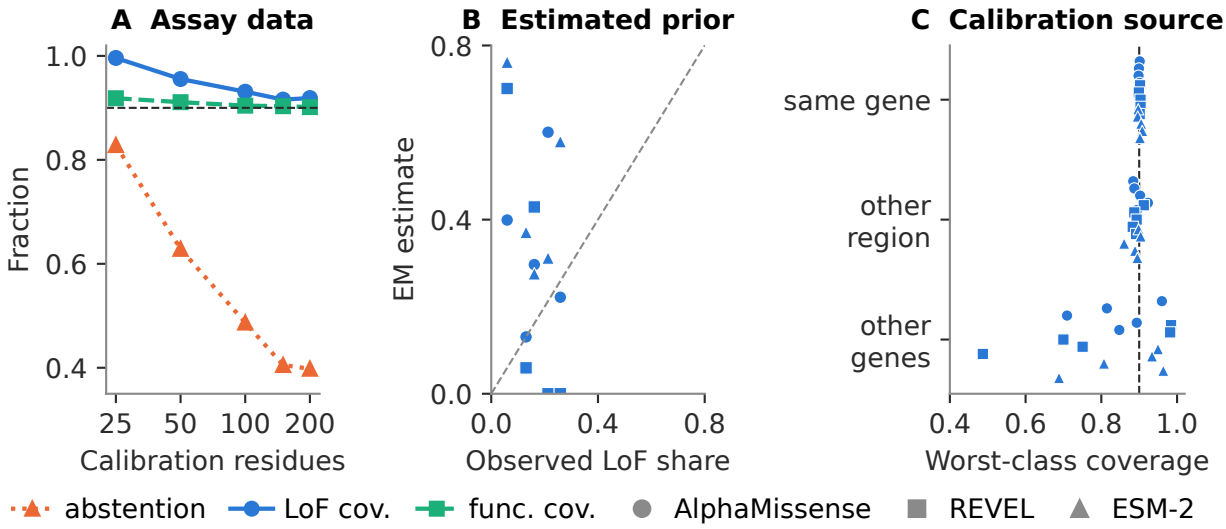


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

S4 Strata, assay-label sensitivity and ClinVar check

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and 0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (Section 4.5). This analysis is

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.916/0.901/0.25	0.952/0.889/0.32	0.971/0.927/0.44
AlphaMissense	BRCA2	0.899/0.903/0.36	0.808/0.921/0.17	0.890/0.965/0.37
AlphaMissense	BAP1	0.909/0.898/0.39	0.977/0.880/0.69	0.942/0.687/0.26
AlphaMissense	RAD51C	0.901/0.905/0.17	0.824/0.929/0.10	0.961/0.973/0.37
AlphaMissense	MSH2	0.915/0.901/0.29	0.972/0.897/0.70	0.956/0.799/0.39
REVEL	BRCA1	0.914/0.902/0.34	0.962/0.883/0.48	0.997/0.999/0.98
REVEL	BRCA2	0.912/0.899/0.42	0.841/0.911/0.24	0.947/0.998/0.68
REVEL	BAP1	0.903/0.904/0.42	0.545/0.997/0.15	0.663/1.000/0.24
REVEL	RAD51C	0.909/0.904/0.27	0.625/0.999/0.20	0.680/1.000/0.23
REVEL	MSH2	0.911/0.901/0.37	0.991/0.896/0.79	0.991/0.900/0.80
ESM-2	BRCA1	0.911/0.897/0.37	0.912/0.893/0.34	0.970/0.975/0.64
ESM-2	BRCA2	0.896/0.903/0.36	0.944/0.894/0.51	0.964/0.958/0.71
ESM-2	BAP1	0.910/0.905/0.47	0.878/0.912/0.36	0.903/0.943/0.46
ESM-2	RAD51C	0.915/0.908/0.34	0.973/0.886/0.50	0.987/0.963/0.71
ESM-2	MSH2	0.920/0.902/0.46	1.000/0.895/0.80	0.993/0.823/0.68

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabelled scores by expectation–maximization (EM) [6] and black-box shift estimation (BBSE) [7]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

an independent re-run with its own random splits, so its main-label rows differ from Table 3 by Monte Carlo error (for example 0.895 against 0.896 in the lowest pair).

Table S4 gives the exploratory check of the calls against ClinVar labels (Section 4.4), per gene and pooled. Only BRCA1, BRCA2 and MSH2 have enough labelled residues for informative bounds; across them the LoF-call lower bounds differ by gene, so the pooled values should not be read as a common likelihood ratio. Some ClinVar classifications already use these assays or predictors as evidence, so this check is not independent clinical validation.

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00

Table S4: Calls judged against ClinVar labels (exploratory). Thresholds ($\alpha = 0.1$) are set on each gene's assay labels; one ClinVar variant with $\geq$2 stars is kept per residue. n: kept P/LP and B/LB variants; calls: LoF (or functional) calls among P/LP / among B/LB variants; L^+, U^-: likelihood-ratio bounds from one-sided Clopper–Pearson endpoints at level 0.025 each (simultaneous level 0.95 per ratio). Pooled rows sum the kept counts over genes, each gene with its own thresholds. BAP1 and RAD51C have too few labelled residues to be informative.

Pred.	Gene	n P/LP	n B/LB	LoF calls	L^+	func. calls	U^-
AM	BRCA1	75	170	56/1	19.6	8/153	0.24
AM	BRCA2	60	210	49/8	9.44	6/164	0.29
AM	BAP1	6	8	5/0	0.97	1/8	1.02
AM	RAD51C	8	3	5/1	0.27	2/2	6.90
AM	MSH2	72	55	50/6	2.58	9/46	0.31
AM	pooled	221	446	165/16	11.9	26/373	0.21
REVEL	BRCA1	75	170	48/10	4.94	8/134	0.28
REVEL	BRCA2	60	210	44/9	7.56	11/182	0.37
REVEL	BAP1	6	8	4/0	0.60	2/8	1.23
REVEL	RAD51C	8	3	6/0	0.49	2/3	2.23
REVEL	MSH2	72	55	43/3	3.14	5/38	0.28
REVEL	pooled	221	446	145/22	7.99	28/365	0.23
ESM-2	BRCA1	75	170	47/1	15.7	10/159	0.26
ESM-2	BRCA2	60	210	33/4	8.66	9/184	0.32
ESM-2	BAP1	6	8	5/0	0.97	1/5	2.62
ESM-2	RAD51C	8	3	5/0	0.35	1/1	62.7
ESM-2	MSH2	72	55	44/3	3.23	3/33	0.25
ESM-2	pooled	221	446	134/8	15.4	24/382	0.19

S5 Implementation

Inputs: ClinVar `variant_summary.txt.gz` of 26 October 2026 (SHA-256 in `data/MANIFEST.txt`); MaveDB score sets `urn:mavedb:00000097-0-2` (BRCA1), `00001279-a-1` (BRCA2), `00000662-0-1` (BAP1), `00000673-0-1` (RAD51C) and `00000050-a-1` (MSH2, positive scores = loss of function); Ensembl VEP REST on GRCh38 (release 116) with the AlphaMissense and REVEL annotations; ESM-2 `esm2_t33_650M_UR50D`. Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint of the two class medians in its gene, divided by the distance between the medians. Expected calibration error uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with fixed seeds (2027 plus a fixed offset

per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository. The Platt map is checked to be increasing (positive slope) every time it is fitted, and every fit in the experiments passed. Certified bounds use the conventions of Section 3: $U^- = \infty$ when $\ell_{\mathrm{TN}} = 0$, an empirical ratio $x/0$ is ∞ for $x > 0$, and $0/0$ is undefined (never a claim); boundary cases are covered by unit tests (`make test`). The VUS-shift LoF shares are residue-weighted for both the calibration sample (one variant per residue) and the VUS (each residue weighted equally, then its variants).

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